

DevOps & Agile

1. What is DevOps?

DevOps is a way of working where development (Dev) and operations (Ops) teams collaborate throughout the entire software lifecycle — from writing code to deploying and maintaining it in production.

Before DevOps, developers would write code and hand it off to a separate ops team to deploy. This caused slow releases, miscommunication, and a lot of blame-shifting when things broke. DevOps removes that wall.

Core ideas:

- **Continuous Integration (CI):** Developers merge code frequently; automated tests run on every change.
- **Continuous Delivery (CD):** Code is always in a deployable state and can be released at any time.
- **Automation:** Testing, building, and deploying are all automated — less manual work, fewer errors.
- **Monitoring:** Systems are constantly watched so issues are caught early.

2. Advantages of DevOps

- **Faster releases:** Teams ship features and fixes much quicker than traditional models.
- **Fewer bugs in production:** Automated testing catches problems before they reach users.
- **Better collaboration:** Dev and Ops share responsibility — no more finger-pointing.
- **Faster recovery:** When something goes wrong, the team can identify and fix it quickly.
- **Scalability:** Infrastructure can be managed as code, making it easy to scale up or replicate environments.
- **Consistent environments:** Dev, test, and production environments are kept the same, reducing 'works on my machine' issues.

3. What is Agile Methodology and What is Scrum?

Agile is a project management approach built around delivering work in small, frequent chunks rather than one big release at the end. The idea is to stay flexible, respond to feedback quickly, and continuously improve.

Instead of planning everything upfront for 6 months and then building, Agile teams plan a little, build a little, review, and repeat.

Key Agile principles:

- Working software over lengthy documentation.
- Customer feedback over following a rigid plan.
- Responding to change over sticking to the original plan.

What is Scrum?

Scrum is the most popular way to implement Agile. It breaks work into fixed time periods called **Sprints** (usually 2 weeks). Each sprint has a goal, and at the end the team delivers something working.

Scrum roles:

- **Product Owner:** Decides what gets built and in what order. Owns the backlog.
- **Scrum Master:** Keeps the team on track, removes blockers, and ensures the process is followed.
- **Development Team:** The people actually building the product.

4. What are the Important Meetings in Agile?

In Scrum, there are four key meetings (called **ceremonies**). Each one has a specific purpose:

Sprint Planning

Held at the start of every sprint. The team picks items from the backlog and decides what they'll complete during the sprint. Sets the sprint goal.

Daily Standup (Daily Scrum)

A short 15-minute meeting every day. Each person answers three questions:

- What did I do yesterday?
- What am I doing today?
- Is anything blocking me?

Sprint Review

Held at the end of the sprint. The team demonstrates what they built to stakeholders and gets feedback. Focuses on the product.

Sprint Retrospective

Also at the end of the sprint, but focuses on the team's process. The team discusses what went well, what didn't, and what to improve next sprint.

5. What are the Advantages of Agile?

- **Faster delivery:** Working features are delivered every few weeks, not months.
- **Flexibility:** Requirements can change mid-project and the team adapts without starting over.
- **Better quality:** Continuous testing and reviews mean issues are caught and fixed early.

- **Customer involvement:** Regular demos and feedback loops mean the product actually matches what the customer wants.
- **Transparency:** Everyone on the team knows what's being worked on and why.
- **Reduced risk:** Small releases mean smaller failures — easier to fix and recover from.